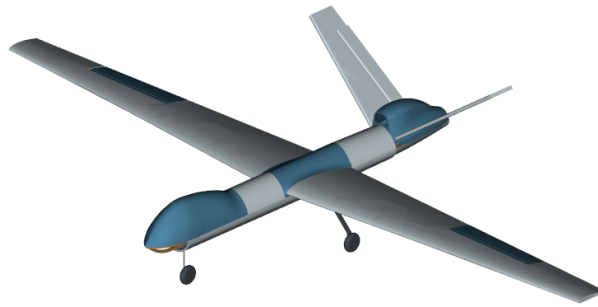


01 / MQ9 airframe

Manufacturing and assembly guide / 2 metre RC model / Revision 1



This guide is for the teammate fabricating and assembling the body. The owner fits and configures the electronics using the separate Electronics Build Guide. Use this revised package as a set; do not mix it with earlier AI conversion files.

Start here

1. Read this guide and the part registers. Print the fit coupons first.
2. Cut plywood and make the printed/metal parts. Mark every part ID.
3. Dry-assemble the complete skeleton and removable wings.
4. Form and fit one representative foam panel, then finish the skins.
5. Deliver the body with access open for electronics, plus the completed inspection record.

Release scope: dimensioned prototype fabrication files with declared corrections and digital checks. Material behaviour, landing loads, completed mass and flight performance have not been physically verified. The shop coupon, dry-fit and ground checks in this guide are required build operations, not claims that they have already passed.

02 / Files, tools and stock

Folder	Use
01_Plywood	Individual CUT profiles, quantity register and revised positioned STEP parts.
02_Foam	Developed knife patterns, forming guide, source mapping and seam evidence.
03_Print	Only selected aircraft print STLs, matching STEP and settings.
04_Hardware	Tube schedule, metal forming/drilling references, motor drawing and coupon.
05_Assembly	Revised assembly STEP and labelled reference illustrations.
06_Verification	Measured checks and declared revisions.
07_Source	Untouched original STEP assembly, individual parts ZIP and transcript.
Build Guide	Instructions, consolidated manual, part atlases and overviews.
Print_Templates	Actual-size A4 cutting and tooling templates. Keep these separate from the reading manual.

Machine work versus bench work

Laser or CNC: plywood CUT layer only. Do not cut labels, sanding guides, sheet edges or reference lines. Print: selected plastic parts. Metalwork: form/drill the gear references. Knife: the confirmed paper-faced XPS foam. Use a laser on XPS only with explicit workshop material/process approval.

Bench work remains essential: identify parts, check fit, bevel and form foam, glue, fit fasteners and linkages, align surfaces and inspect moving clearances. A machine export does not assemble the aircraft.

Stock and setup

Use aircraft/model plywood of the indicated thickness, including 2.5 mm nose doublers. Use the supplied 5 mm FliteBoard Pro, 1000 x 600 mm, nominal 400 g/m². Keep balsa optional. Use straight matched sliding carbon tubes as specified on page 6; 5 mm solid rods are not replacements.

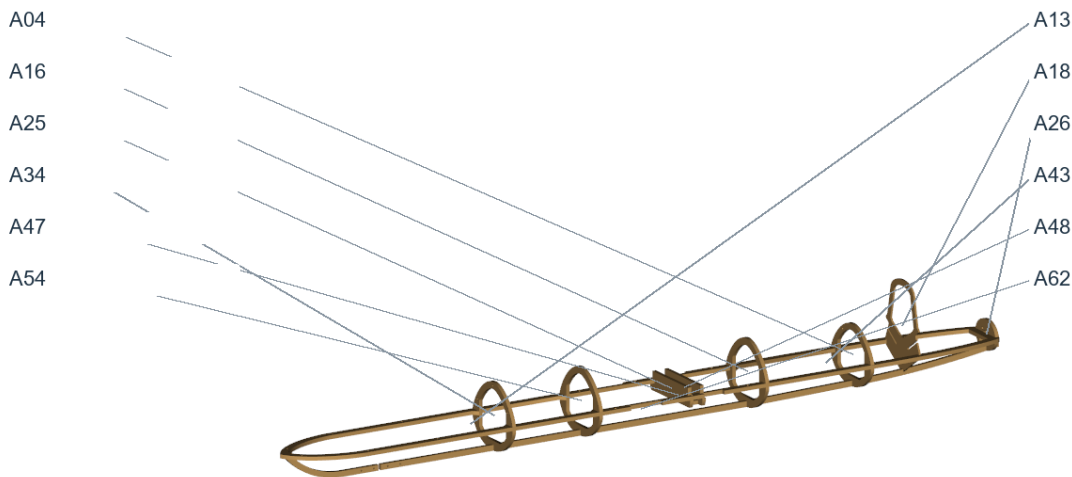
The reference plywood sheets are 1220 x 2440 mm; the occupied area is much smaller. Shops may reneat at 100% with quantities and grain directions preserved. Long spars/spine/doublers need face grain along their length. Never scale a part to fit a machine bed. Keep material, part IDs and sheet thickness together.

03 / Parts and buying list

Item	Quantity / specification
Plywood	31 cut pieces from 22 profiles: 24 nominal 5 mm, 4 at 3 mm, 3 at 2.5 mm. Two outer ribs are 4.990 mm CAD thickness; finish 5 mm stock to fit.
Foam	5 mm FliteBoard Pro. Follow per-pattern quantities; moving tail controls use 2 layers each. Ten supplied sheets are starting stock, not ten installed sheets.
Printed aircraft parts	11 individual STLs. One fit coupon is additional and is not installed.
Carbon	Front 8/6 mm joiner 406 mm x 1; rear 8/6 mm 390 mm x 1; matched 10 mm OD/8.2 mm nominal ID sleeves 199.35 mm x 2.
Wheels and gear	3 lightweight 50.8 mm wheels, 3 mm bores; formed main gear and 3 mm spring-wire nose gear; steering arm and shaft collars.
Wing root retention	6 M2 bolts, start 20 mm long; 12 washers and 6 locking nuts. Verify final stack.
Nose bypass	2 M2x 40-45 inner bolts +2 M2x 16 outer bolts; 8 washers, 4 locking nuts. Select actual length from stack.
Main gear fastening	2 M3x 16 through-bolts, large washers, 2 locking nuts; 2 M3 axle bolts, nominal 25 mm plus spacers/nuts.
Motor fastening	4 M3 screws, matching washers; determine screw length from 5 mm ply and actual safe motor thread depth.
Nose bearing	4 M3 bolts through bearing/mount, washers and locking nuts; start 16-20 mm and verify stack/clearance.
Controls	5 servos: 2 ailerons, 2 tail, 1 nose steering. 4 control horns, 1 steering arm, 5 pushrods with 10 clevises/ball links; servo screws.
Hinge pins	2 smooth 3 mm rods; 285 mm initial length, trim after fitting retainers.
Adhesives / covering	Wood-compatible epoxy/wood adhesive; XPS-compatible foam adhesive; reinforced tape and paper seam straps. Qualify on offcuts.

Stock fasteners are starting procurement lengths where the real nut, washer and servo horn affect the stack. Trim/select them at dry fit. Add spare washers, locking nuts, cable ties and servo extension leads. Do not use an unspecified screw through a motor base or into foam.

04 / Fuselage skeleton



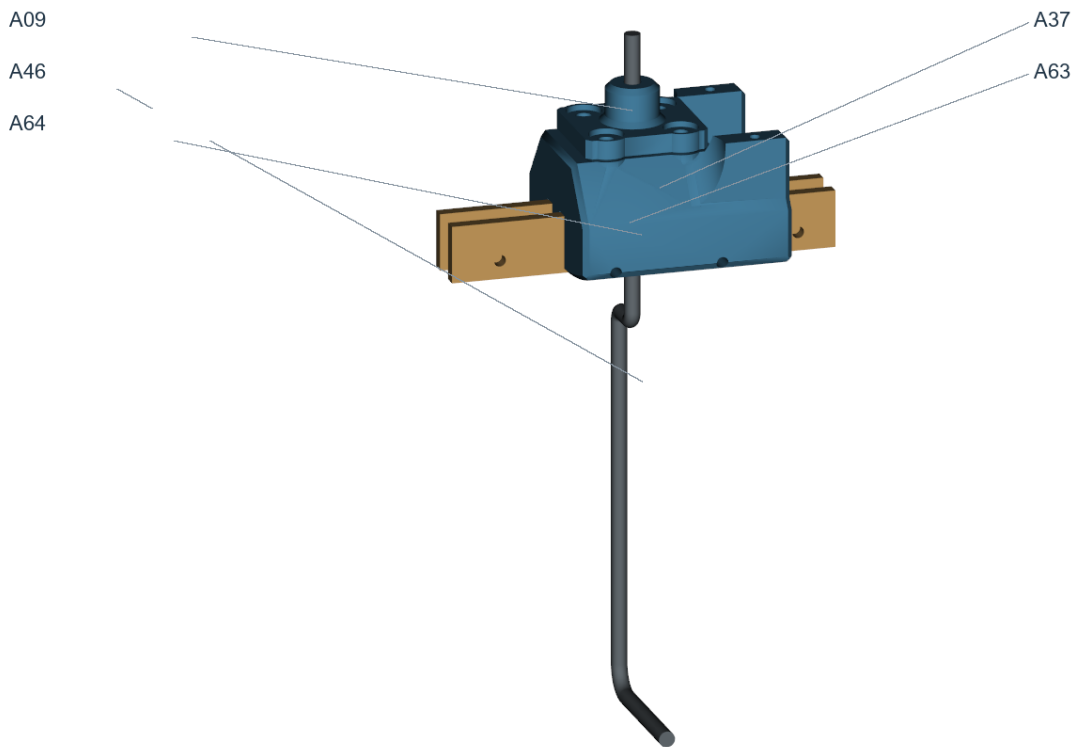
CAD coordinates: X is spanwise, Y increases from nose to tail, Z is up. Nose tip is approximately $Y = -439.65$ mm. The revised assembly STEP is the placement reference. The source typo `Notor_Mount` is the motor mount A26.

1. Lay A62F and A62R on a flat jig along the centreline; first make the nose bypass on page 5. Do not glue them end-to-end across the shaft gap.
2. Fit forward longitudinal member A13 and aft member A43, then former A34 at $Y = -130$, A54 at $Y = -5$, A16 at $Y = 233.65$ and A04 at $Y = 384.65$. Slots determine final seating.
3. Fit the main-gear box A25/A48 with bottom plate A47. Fit tail bulkhead A18 and revised motor mount A26.
4. Check squareness from the assembly axes, equal left/right former extents, and that all tabs seat without forcing. Test fit the printed nose, centre and tail fairings before bonding.

Bond wood joints after the full dry assembly fits. Use modest structural fillets at gear, wing-root and motor load paths; remove excess adhesive rather than adding mass indiscriminately. Keep battery/avionics access open. Stock thickness and laser kerf control the actual fit; resolve a tight joint by the coupon/process setting, not by scaling the DXF.

Revision: overlapping A40/A62 source spines have been consolidated. There is no separate A40 piece to install. The nose shaft requires the intentional A62F/R split and bolted side doublers.

05 / Nose gear bypass



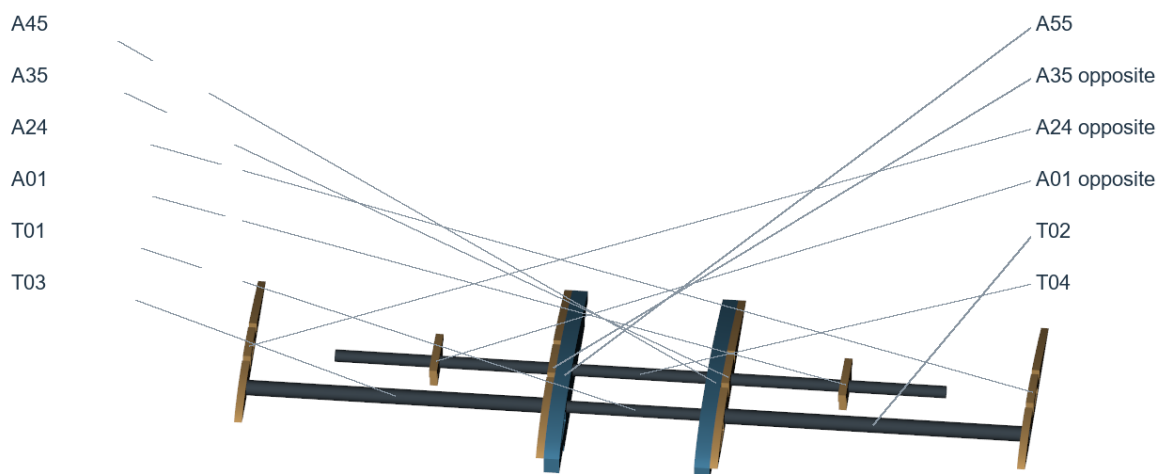
The original 3 mm steering shaft passed through the thin centre spine. The revised joint routes the connection through two continuous 2.5 mm plywood cheek plates A63/A64 and four M2 bolts. A37 has matching recessed pockets. Each cheek is 72 x 10.7 mm; face grain runs along its length.

1. Place A62F/R collinear with the 3.4 mm shaft gap centred on $Y=-293.65$.
2. Fit one cheek on each side of the 5 mm spine. Dry-fit them into the revised A37 pockets.
3. Install four transverse M2 bolts at $Y=-320, -302, -282, -264$ mm, $Z=-37.964$ mm. Inner bolts pass through the mount; outer bolts clamp the wood stack. Use washers and locking nuts.
4. Fit A09 and the extended 3 mm wire. Rotate the shaft fully. Bond wood interfaces only after the assembly stays straight and turns freely; keep glue out of the shaft passage.
5. Tighten enough to seat the stack without crushing plywood/plastic. Recheck after cure.

The corrected wire extends its upper shaft to $Z=+1$ mm, through the bearing and steering-arm region. Fit an actual 3 mm clamp arm/collar and trim only after engagement is established. Do not copy the short, disconnected shaft in the original assembly. The shaft must not rely on glue for retention.

Ground inspection: support the body, steer both directions, move the wheel through its allowed deflection and check the bypass, mount and skin opening together. Inspect for cracks, loose bolt bearing and rubbing before flight. No landing-load rating is asserted by the digital fit check.

06 / Carbon joiners and retention



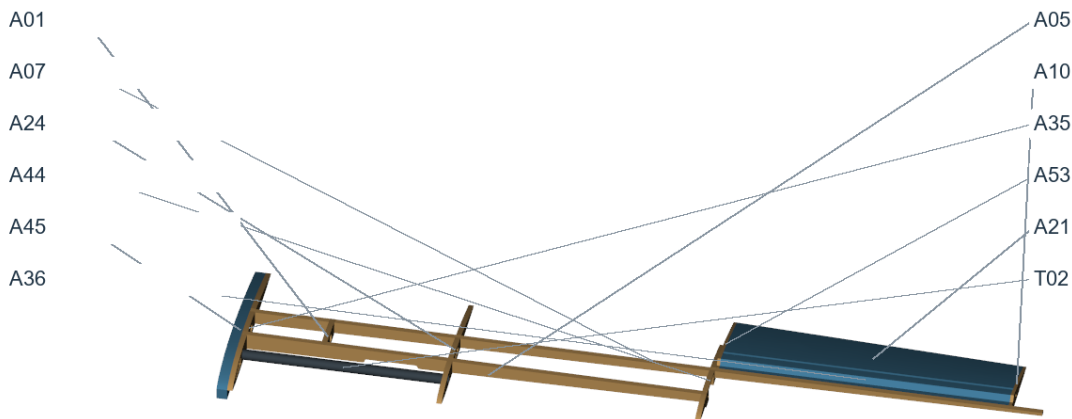
Part	Section / cut length	Position and purpose
T01	8 OD /6 ID; 406 mm	CentreX=-203..203. Front joiner axisY60.454, Z15.632.
T02/T03	10 OD /8.2 nominal ID; 199.35 mm each	RightX53..252.35; leftmirror. Bond sleeves into wing root/middle ribs.
T04	8 OD /6 ID; 390 mm	CentreX=-195..195. Rear axisY100.500, Z15.070.

Buy or have the shop make a matched slip-fit pair. Nominal 8.2 mm sleeve ID gives 0.2 mm diametral clearance around 8 mm joiner; actual tube tolerances govern fit. Do not force unmatched stock or silently sand away structural wall thickness. Plywood sleeve holes are 10.2 mm; root/rear support holes 8.2 mm.

Dry-fit both wings on the centred tubes before bonding sleeves. Front insertion is approximately 150 mm each side into the wing sleeve. Keep adhesive off the sliding joiner. Retain each removable wing using the three aligned M2 root fasteners. Check there is no spanwise escape, twist or hinge interference. Mark insertion positions on tubes.

This replaces overlapping source tube references and recentres the rear tube. Do not cut the source A28 stub or use source A33 as a second 10 mm tube. Your 5 mm solid carbon rods are spare material, not the wing-joiner specification.

07 / Wing frames and ailerons



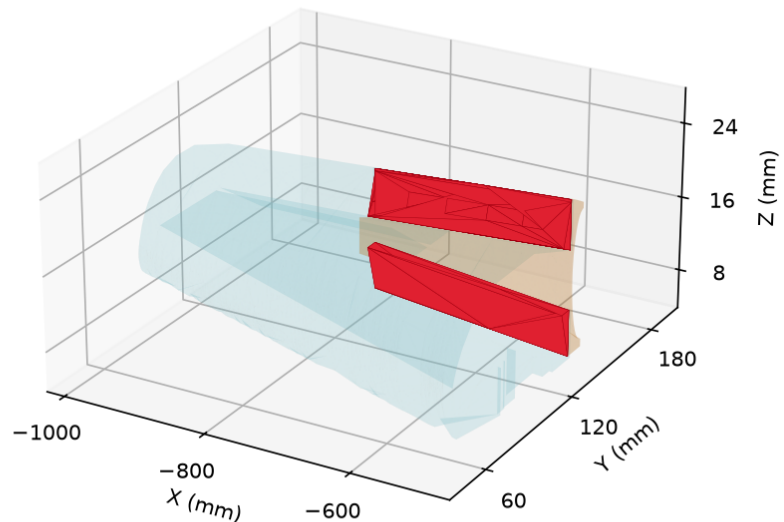
Build one right wing, then the corresponding left-hand mirror. Plywood through-cut profiles are reused at the listed quantity; assembled STEP files show the mirrored positions. Do not build two right wings.

1. Fit inner rib A35, middle rib A24 and outer rib A44 to front spar A05 and rear spar A07. Fit rear joiner support A01 and the front sleeve. Match each rib slot, not a guessed spacing.
2. On a flat jig, check leading/trailing edges and equal incidence before bonding. Finish A24/A44 edges to the marked noncut sanding guide where needed; preserve revised bore centres.
3. Install the aileron end plates A53/A10, hinge rail A36 and aileron A21 on the right; use A23/A11 on the left. Dry-fit smooth 3 mm pins and retain them mechanically, leaving about 1 mm axial freedom.
4. Install each servo with its manufacturer screws and a serviceable mounting pad. Transfer actual servo mounting holes after confirming its size; the CAD servo is a reference envelope.
5. Fit the control horn and neutral pushrod before closing the skin. Check travel with no scraping, stiffness or exposed sharp pin end.

Do not glue ailerons to the fixed rail. Print a moving-fit sample first; finish holes/hinge interfaces to the actual pin with the outside dimensions unchanged. Do not solve a tight hinge by scaling the entire STL.

08 / Foam skins and seams

Wing hinge inner core | red = CAD interference to relieve



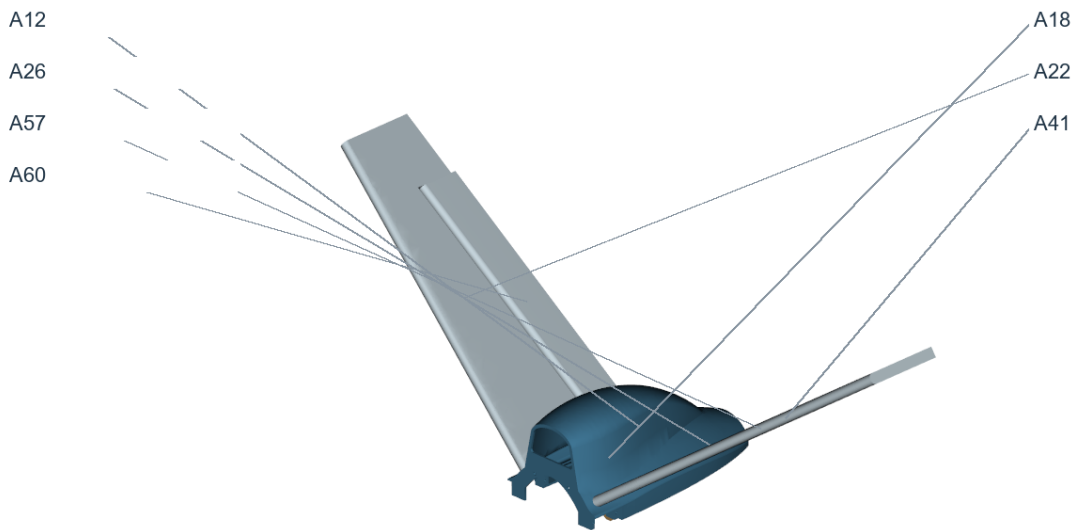
Blue: A27 foam. Tan: A23. Red: trim contact region. View cropped; axes not equal scale.

Use the developed F-series patterns and their forming guide. They follow the curved CAD surface. Print at 100%, check scale in both directions, and assemble paper tiles using the common marks/window coordinates and 10 mm overlaps.

1. Mark each outside face and part ID. Knife-cut the finished skin edge. Moving-tail blanks have 5 mm sacrificial shaping stock. F23/F24 include a 1 mm PAPER-ONLY lower seam lip: remove its core and trim/fold it against the mating lip as shown in the foam guide. Other skin CUT lines have zero added lap/trim.
2. Peel inside paper only in the curved forming zones. Keep outside paper intact as the pattern reference. Gradually relieve/compress inside foam and dry-form over the real frame; do not stretch a short paper edge into place.
3. Seat the wing from its leading-edge region toward the open/trailing seam. Shave local inner foam for the printed hinge rail and rib contact patches. Do not cut away the outside paper to hide an inner interference.
4. Bevel inside seam edges to their actual meeting angle. Butt outside paper edges and add 15 mm internal paper straps, about 7.5 mm per side. Reinforce exposed seams with suitable light tape.
5. Dry-fit an entire wing and its moving surface before gluing progressively along the frame. Hold alignment while adhesive cures.

Peeled board loses stiffness and can warp. Use symmetric restraint and prompt bonding to the frame; reinforce seams and relief boundaries without reintroducing an interference. Make the forming/adhesive coupon before cutting the full board set. Use XPS-compatible glue; ordinary CA must not touch exposed XPS unless its manufacturer explicitly approves it.

09 / Fuselage covering and tail



Fit printed nose A59, centre fairing A06 and rear/tail fairing A12 to the dry skeleton. A12 includes the revised bulkhead pocket. Keep a removable battery/electronics access panel rather than permanently closing every cover before the owner has balanced the aircraft.

Match F-series fuselage panels to their source-face diagrams. Install lower forward, lower middle and upper panels in their numbered groups. The rear underside is divided into narrow developed bands: assemble them in parameter/number order and support each butt seam internally. Shave inner-core contact patches against the actual longitudinal wood and spine before bonding.

Cut the main-gear and nose-wire exit only after tracing the installed gear. Start with 0.5 mm around the static leg, then check the required deflection/steering clearance on the ground. The foam must not carry gear loads. Reinforce the paper around the opening.

For each fixed V-tail, use the removable tail-forming collars supplied with the foam tooling PDF to check the exterior at the marked stations; they are tools, not aircraft parts. Form its paired skins/root collar and test slot engagement in A12. The separate control is 10 mm thick: laminate two 5 mm blanks, transfer the finished reference outline and shape the long edges to R5 using the STEP/end reference. Discard the sacrificial 5 mm perimeter allowance.

Make tape hinges that preserve the CAD hinge line and permit free symmetric movement. Fit two tail servos, horns and pushrods with access for adjustment. Do not bridge a hinge with a rigid seam strap or glue. Fit both tails together and compare symmetry to the revised assembly before the joint cures.

10 / Printing and metalwork

Print group	Starting process settings, then qualify on a coupon
A06/A12/A59 shells	LW-PLA preferred for mass. If using available PLA+, start 0.2 mm layers, 2 perimeters and 3-5% infill; record slicer mass and weigh the finished part. Support internal overhangs without blocking sockets.
A11/A21 ailerons; A23/A36 rails	Start 3 perimeters, 10% infill; orient long axis on the bed with supports kept out of hinge bores. Use smooth pin sample and full travel check.
A09/A37/A45/A55 functional parts	Prefer a shop material/process suited to load and temperature; available PLA+ requires inspection/testing. Start 4 perimeters, 30% infill. Orient so tube/bolt loads do not simply split layer interfaces. Do not approve from infill percentage alone.

STLs are millimetres and translated to positive coordinates without scaling. Shop selects final orientation and support strategy. Some parts exceed common 220 mm beds; use a larger printer or have an explicit joint designed. Do not cut a structural STL into arbitrary pieces or scale it down.

Clean support material completely, check every passage and inspect layer bonding. Compare caliper dimensions against the print manifest. Install no cracked, warped or incompletely bonded root bracket. Record total print mass before the owner decides battery placement.

Landing gear

A02 is a formed metal strip reference, 12.7 mm wide by 3.175 mm thick. Its DXF is the formed front view, not a flat laser blank. The metalwork shop chooses an appropriate bendable alloy/temper and real bend radii, preserving the mounting and axle stations. It must calculate its own tooling bend allowance. Do not attempt ideal zero-radius bends from the CAD.

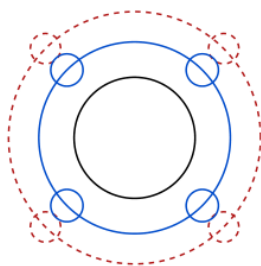
Use the revised two 3.2 mm mounting holes with M3 through-bolts, washers and locking nuts into A47. Fit the three purchased 50.8 mm wheels. Form the 3 mm spring-wire nose leg to the extended STEP reference. These gear references require load/deflection inspection on the completed aircraft; geometry alone does not rate them for a landing.

11 / Motor mount and handoff to electronics

A26 motor mount - declared drilling correction

Source: 4x3mm on 25mm PCD. Revised: 4x3.2mm on 19mm PCD.

Physical Emax motor check required before drilling; drawing below at 2:1.



Red dashed: obsolete 25mm PCD - DO NOT DRILL

Blue: 19mm PCD - four 3.2mm holes

Central 12mm hole unchanged. Minimum radial wood web 1.9mm; use washers.

Revised A26 has four 3.2 mm clearance holes on a 19 mm pitch circle for the Emax ECO II2807. Hole centres are 13.435 mm apart on the square diagonal orientation shown. This replaces the original 25 mm circle. Keep the 12 mm centre opening.

Place the physical motor on the template before cutting/drilling production material. The official drawing is included because marketing wording can be ambiguous. Fit washers and screws with measured safe thread engagement; an overlong screw can reach the windings. Route wires without contact with the rotating bell.

The owned 9-inch propeller is not approved by this fabrication package. The Emax source recommends 6-7-inch propellers, and no verified current/thrust test of the owned 9-inch combination has been supplied. Leave the propeller off during assembly/calibration. The electronics owner must qualify motor/prop/ESC load and ground clearance before flight.

Leave these accessible

Battery tray and straps; FC mounting area; ESC cooling/wiring path; receiver antennas; GPS/compass mount; airspeed tube routing; wing connectors; servo horns; steering joint and gear fasteners. Fit pull strings before skins close. Do not fix the battery position from appearance alone.

Deliver the body with every loose fastener bag labelled, the revised assembly reference, the actual part weights and notes on every fitted/sanded area. The owner uses the separate electronics PDF for wiring, calibration and powered tests.

12 / Checks before the body is signed off

Check	Builder records
Scale and coupons	100 x 20 mm rectangle measured; actual ply/foam thickness; kerf/slot result; printer coupon result.
Structure	All 31 wood pieces accounted for; no source A40 duplicate; bypass bolts installed; no split grain/cracks; aligned centreline.
Wing attachment	Both wings fit simultaneously; matched sleeve fit; centred tubes; six M2 retainers; no spanwise escape.
Controls	Two ailerons, two tail controls and nose steering free; hinge pins retained; horns secured; no glue in moving gaps.
Skins	Full surface dry-fit; seams meet without forced stretching; core relief/gear exits complete; outside paper sound; no wing twist.
Gear and motor	Positive gear fasteners; free wheels; free steering; motor pattern/screw engagement checked; ground clearance examined.
Mass and access	Total body and individual print masses recorded; battery/FC access open; wires can be serviced.
Outstanding evidence	Material forming, adhesive/print strength, gear loading, propulsion and flight checks must be recorded physically.

Balance reference for the owner

The geometry-derived quarter-MAC station is approximately 522 mm aft of the nose tip, or 22.2 mm aft of the front joiner axis. It is a provisional setup mark, not an aerodynamic or flight-approved CG range. Exposed wing area is 0.30445 m² and mean aerodynamic chord 169.83 mm. Mark the datum on the body and leave battery travel available.

Check balance only in complete flight configuration. Moving the battery changes CG by battery mass / total aircraft mass times the movement. Static geometry does not establish elevator authority, stability margin or take-off capability. An experienced fixed-wing pilot must assess the completed configuration before flight.

Digital evidence

The package includes native profile checks, revised valid solids, printable mesh checks, surface development/seam evidence and numerical PDF scale/registration checks. The Verification folder and README state the scope and remaining physical operations. Original CAD, drawings and transcript remain traceable in Source. All revisions are listed; no global scaling was used.

Sources: supplied Onshape STEP/transcript; official Emax mounting drawing; manufacturer material information. Electronics sources and model-specific wiring references are in the owner guide. Illustrations show actual source/revised CAD; colours identify material groups, not final paint.